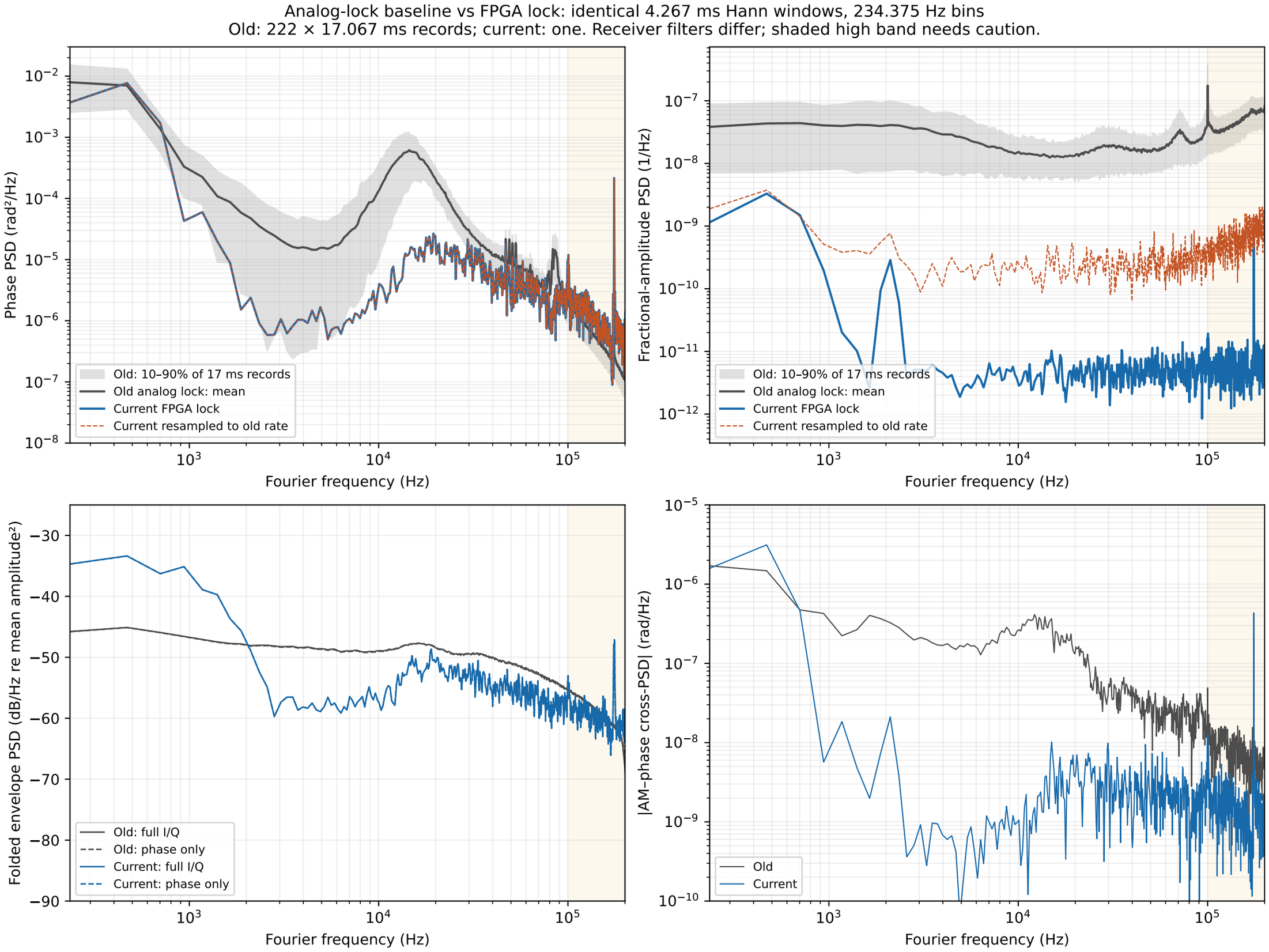

Analog-lock baseline vs FPGA lock: identical 4.267 ms Hann windows, 234.375 Hz bins
Old: 222 × 17.067 ms records; current: one. Receiver filters differ; shaded high band needs caution.
Phase PSD (rad²/Hz)
Fourier frequency (Hz)
Old: 10–90% of 17 ms records
Old analog lock: mean
Current FPGA lock
Current resampled to old rate
Fractional-amplitude PSD (1/Hz)
Fourier frequency (Hz)
Old: 10–90% of 17 ms records
Old analog lock: mean
Current FPGA lock
Current resampled to old rate
Folded envelope PSD (dB/Hz re mean amplitude²)
Fourier frequency (Hz)
Old: full I/Q
Old: phase only
Current: full I/Q
Current: phase only
|AM–phase cross-PSD| (rad/Hz)
Fourier frequency (Hz)
Old
Current